

IMAN AFYOUNI

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SUMMARY

M.Sc. in Materials Science specializing in **energy storage materials** with an and a B.Sc. in Mechanical Engineering. Brings 4+ years of combined **R&D** and **manufacturing** experience, including hands-on expertise in material synthesis, electrochemical testing, and advanced characterization (SEM/EDS, XRD). Track record of improving material performance through systematic process optimization and data-driven experimentation. Seeking to apply deep materials expertise to process engineering or R&D roles in the energy, advanced manufacturing, or specialty materials industries.

WORK EXPERIENCE

Lab Research Assistant Aug 2024 – Dec 2025

Pittsburg State University – Department of Materials Science | Pittsburg, KS

- Optimized active material (Cathode-anode) synthesis for **zinc-ion** and **lithium-ion batteries**, improving specific capacity and extending cycle life. presented findings at the American Chemical Society (ACS) national conference.
- Evaluated **coating** strategies for **manganese-based** and **vanadium-based** active materials, optimizing particle size distribution and enhancing material characterization.
- Employed **XRD** analysis to study structural changes as a function of temperature and hydrothermal duration.

Lab Research Assistant May 2023 – Jan 2024

University of Houston – Department of Materials Science | Houston, TX

- Increased supercapacitor specific capacitance through **electrolyte optimization** and improved rate capability by 20% at high current density.
- Monitored charge storage behavior and **long-term cycling** stability to evaluate supercapacitor lifetime performance.

Quality Assurance & R&D Coordinator Aug 2020 – Aug 2022

EFA Faraz Avina Co. | Iran. (Mid-sized industrial equipment manufacturer specializing in production machinery.)

- Conducted root cause analysis on recurring production defects, implementing corrective actions that improved product yield and reduced scrap rate.
- Co-designed an industrial model used in manufacturing equipment; produced detailed 2D/3D drawings in **AutoCAD** and **SolidWorks** to meet manufacturing and tolerance specifications.
- Designed dies produced via **CNC** machining, coordinating with production teams to ensure dimensional accuracy and on-time delivery.
- Coordinated cross-functional teams across **R&D**, engineering, and production to resolve process deviations and support timely product launches.

Quality Assurance & R&D Coordinator Jun 2016 – Mar 2019

Sepanta Instrument Engineering Co. | Iran. (Engineering specializes in precision instrumentation.)

- Served as an industrial designer for CNC-machined dies, developing detailed drawings and specifications used directly in production.
- Coordinated QC, R&D, and production teams to resolve process deviations and maintain on-time project delivery.
- Supported continuous improvement initiatives, contributing to documented process changes that increased manufacturing consistency.

SKILLS

Characterization & Testing: SEM/EDS, XRD (phase analysis, structural characterization), EIS, CV, GCD, C-rate testing, long-term cycling stability, particle size analysis, electrode slurry preparation, and mechanical testing.

Materials & Process: Material synthesis, electrolyte optimization, coating strategies, process optimization, root cause analysis, quality assurance, die design, CNC machining coordination, production process analysis

Software & Data: OriginLab, Match, VESTA, VersaStudio, AutoCAD (2D/3D), SolidWorks, MATLAB, Microsoft Office (Word, Excel, PowerPoint); working knowledge of statistical data fitting and visualization

Professional: Technical report writing, cross-functional team coordination, Design of Experiments (DOE), project planning, scientific presentations.

EDUCATION

M.Sc. in Materials Science (GPA: 3.4)

B.Sc. in Mechanical Engineering (Manufacturing & Production Technology) (GPA: 3.5)